Translation Is Not Representation

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Main result. Bias scores from cross-lingual benchmarks can remain meaningful even when translated prompts route through an English hub.
Contributions. English-hub routing beyond CKA; routing-behavior dissociation; audit protocol.
Scope. 4 matched 13B models; 41 layers; 2,142 English-Japanese and 261 English-Korean pairs; CKA, lenses, patching.

1. Problem: Translation Assumes Comparability

Translated bias benchmarks score stereotype preference after translating an English BBQ item into another language. This treats translation as construct preservation.

Question. Is a target-language bias gap behavior, or English-hub routing?

Model	Training regime
Llama-2	English-centric base
Swallow	Llama-2 + Japanese adaptation
koen	Llama-2 + Korean adaptation
LLM-jp	balanced Japanese/English pretraining

Matched pairs fix architecture and initialization; changes mostly reflect adaptation.

2. Motivation: Same Event, Different Continuation

BBQ context English → neutral ice / drinks

JBBQ context Japanese → *nihonshu* rice wine

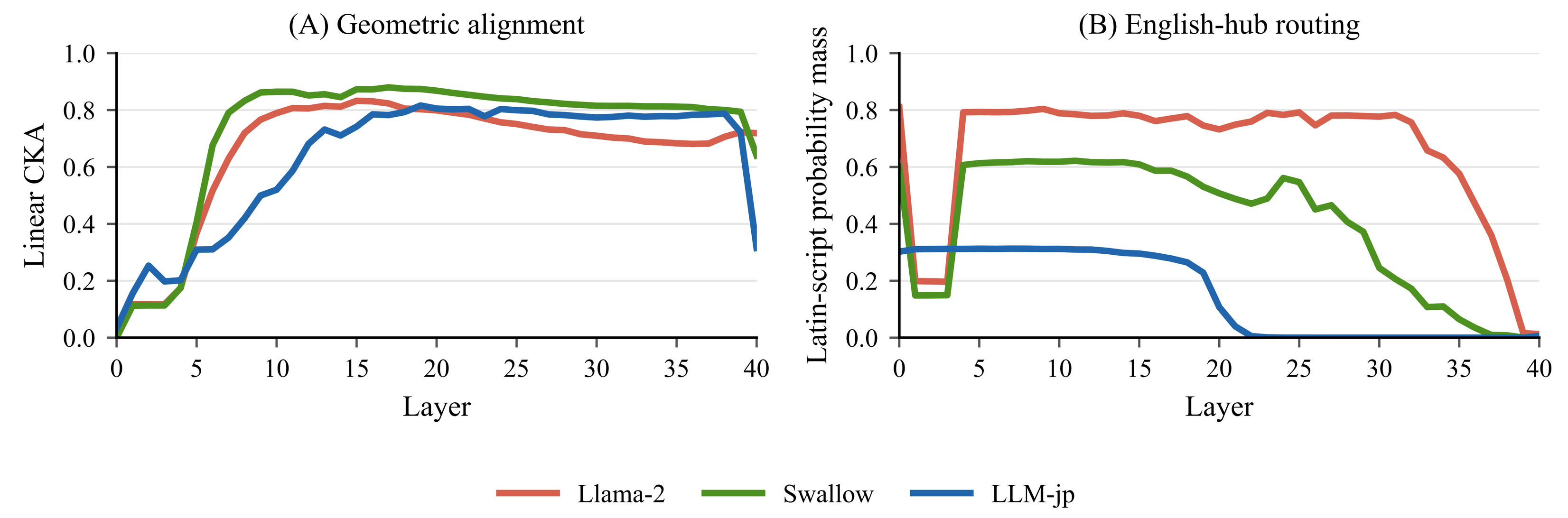
A BBQ age item continued in English yields a neutral exchange about ice; the faithful Japanese (JBBQ) translation instead centers on *nihonshu* (rice wine). The event is preserved, yet the model activates different cultural associations, so a faithful translation need not preserve the measured construct.

3. Method: Four Checks for Comparability

CKA	geometric alignment
Script-mass lens	predicted token script
Binned lens JSD	semantic divergence
Hub patching	causal transplant test

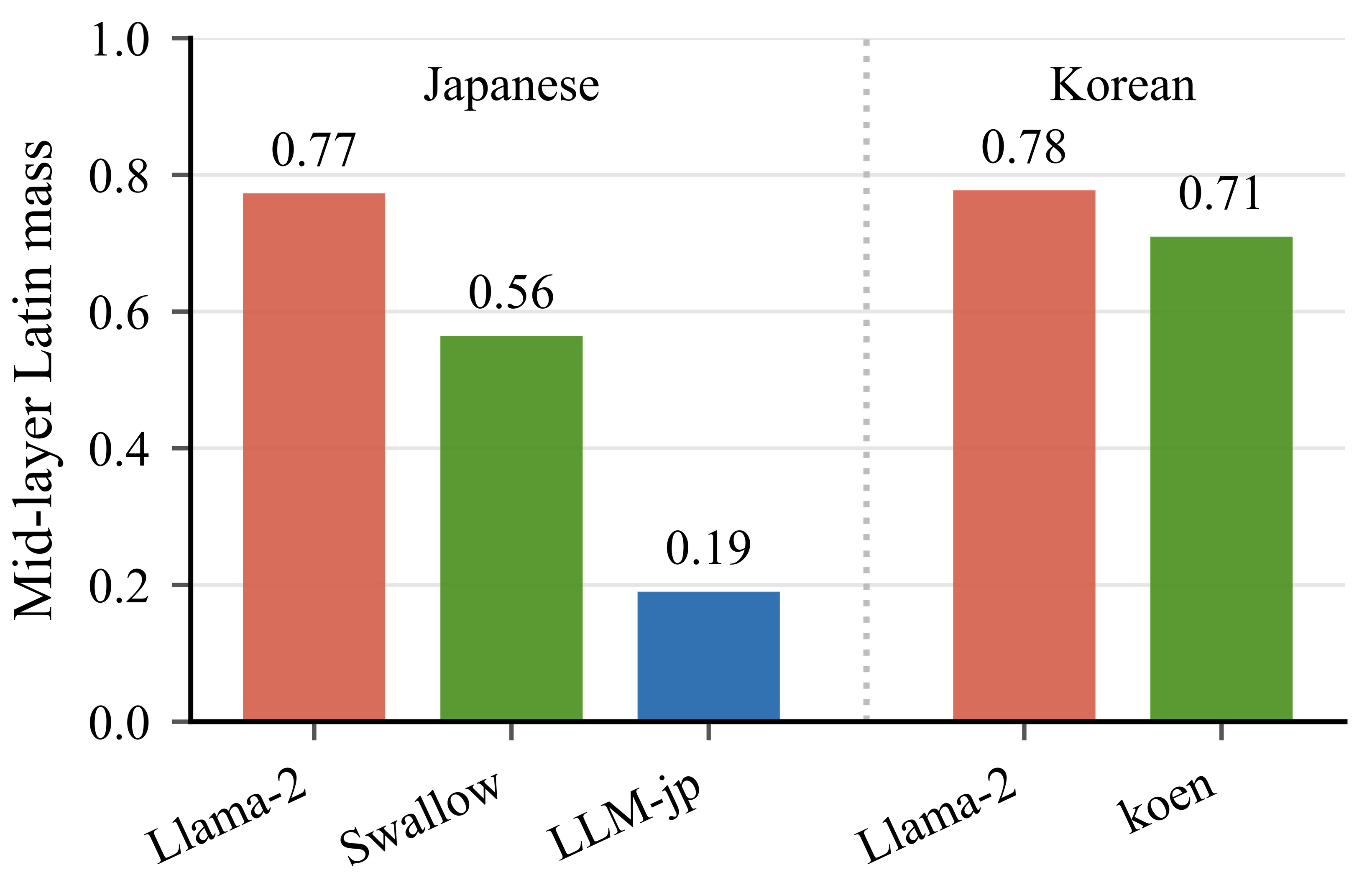
Comparability has two levels: representations should occupy similar regions (geometric, CKA) *and* imply similar predictions (functional, the lenses). Geometric similarity alone is insufficient. Representations use context-only hidden states; behavior uses the full question-answer item.

4. Routing: CKA Hides English-Hub Routing



CKA between English and Japanese is high for every model (peaks 0.83/0.88/0.82), and the English-centric Llama-2 is no lower than balanced LLM-jp. Yet the script-mass lens shows Llama-2 predicting English-script tokens through the middle layers (Latin mass ≈ 0.78), while LLM-jp predicts Japanese from the start. Geometric convergence hides the routing.

5. Adaptation Reduces Hub Routing

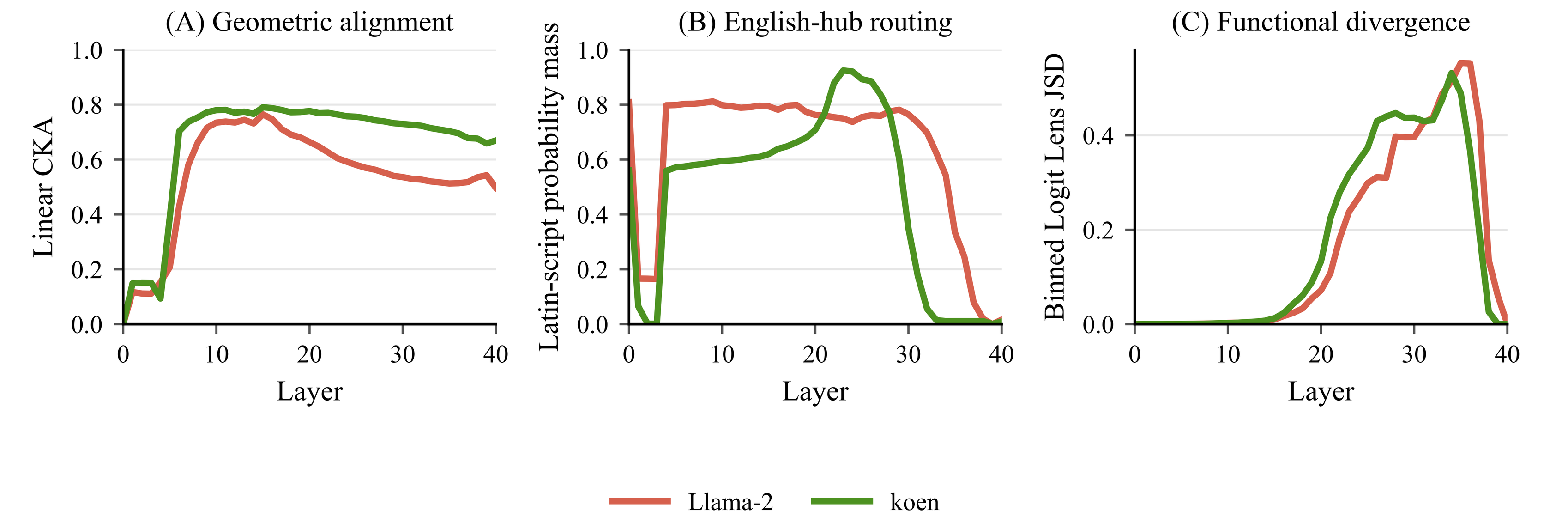


Matched pair (Llama-2→Swallow, same initialization and architecture): the mid-layer Latin-mass drop 0.77 → 0.56 is attributable to Japanese adaptation, not to scale or architecture. Korean replicates (0.78 → 0.71), and balanced bilingual training pushes it furthest down, to 0.19.

Paper

Code

6. Replication: English-Korean Routing



On Korean, Llama-2 routes through English as much as for Japanese (0.78 vs. 0.77); Korean adaptation (koen) reduces it to 0.71, raises Hangul-script mass 0.01 → 0.16, and sustains higher CKA into the upper layers. The same diagnostic split reproduces.

7. Behavior Is Language-Specific

Japanese behavior

Model	EN	JA	Diff
Llama-2	0.487	0.344	+0.142
Swallow	0.496	0.352	+0.144
LLM-jp	0.473	0.343	+0.130

Korean behavior

Model	EN	KO	Diff
Llama-2	0.562	0.508	+0.053
koen	0.507	0.541	-0.034

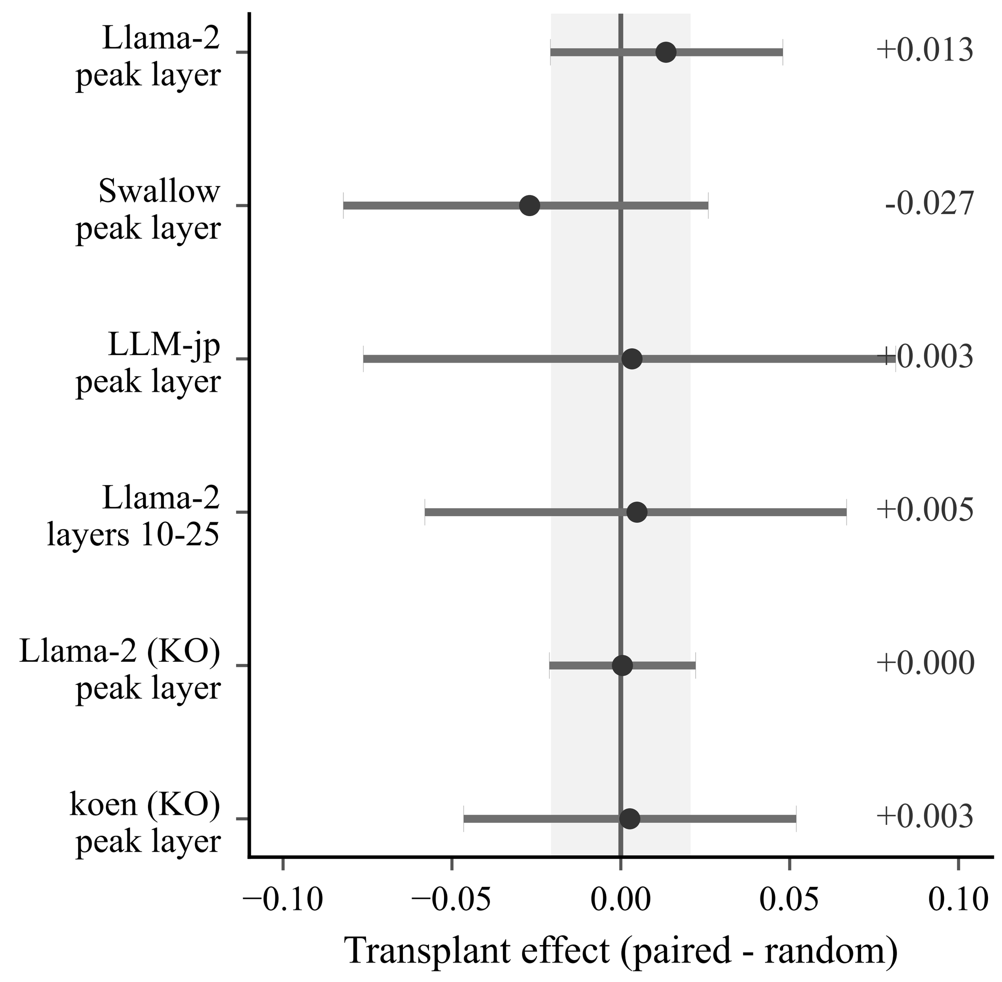
P_{ster} in ambiguous items (chance = 0.33). Japanese gaps are all significant (+0.13 to +0.14) with Japanese near chance; the Korean base gap is small (+0.053) and vanishes after adaptation. The asymmetry is language-specific, not universal.

8. Japanese Gap Varies by Category

Category	EN	JA	Diff
Age	0.491	0.305	+0.186*
Disability	0.477	0.305	+0.172*
Gender identity	0.577	0.446	+0.132*
Physical appearance	0.462	0.463	-0.001
Sexual orientation	0.444	0.271	+0.173*

* bootstrap 95% CI excludes zero. Age, disability, and sexual orientation show the largest gaps (+0.17 to +0.19); physical appearance is the exception, equal across both languages (≈ 0.46).

9. Causal Test: Hub Routing Does Not Transplant Bias



We inject the English hub state at the CKA-peak layer and across the full hub band (layers 10–25). The multi-layer patch perturbs outputs (−0.03 on average) but *non-specifically*: paired and random English states are indistinguishable. Korean gives the same null (transplant +0.000, +0.003). The hub representation is present but carries no item-specific bias.

10. Takeaway and Audit Protocol

Representations: translated prompts are not necessarily processed in comparable spaces.

Behavior: per-language bias scores remain meaningful; patching finds no transplant of English bias.

1. Check tokenizer feasibility.
2. Pair CKA with a script-level lens.
3. Interpret scores as behavior.
4. Filter translations with surface-form chrF.

Step 4 is not vacuous on our data: a chrF back-translation filter shrinks the base model's English/Korean gap from +0.053 to +0.017, flagging low-fidelity items a representation-space cosine would miss.

References: Parrish et al. 2022; Yanaka et al. 2025; Jin et al. 2024; Wendler et al. 2024; Kornblith et al. 2019.